1. Dataset Overview

1.1 Data Source and Construction

The dataset used in this project is constructed from macroeconomic indicators obtained from the Federal Reserve Economic Data (FRED) database. The selected variables represent key dimensions of the macroeconomy, including financial conditions, labour market dynamics, and real economic activity.

The main indicators included in the dataset are:

- **USRECD**: NBER recession indicator (binary variable indicating recession periods)
- **T10Y2Y**: Yield curve spread (10-year Treasury yield minus 2-year Treasury yield)
- **UNRATE**: Unemployment rate
- **INDPRO**: Industrial production index
- **CPIAUCSL**: Consumer price index
- **TEDRATE**: TED spread (measure of financial stress)
- **BAA10Y**: Corporate credit spread
- **FEDFUNDS**: Federal funds rate (monetary policy indicator)

These variables capture different aspects of the macroeconomic environment and are widely used in empirical studies of business cycle dynamics and recession prediction.

1.2 Raw vs Transformed Dataset

Two versions of the dataset were produced by the data infrastructure pipeline:

- **Raw dataset**
- **Transformed dataset**

The transformed dataset applies several econometric transformations to ensure stationarity, including:

- first differencing of trending variables
- log year-over-year transformations
- removal of level variables

While such transformations are appropriate for econometric modelling, they remove important information about the **economic level and structural relationships between variables**.

Since the goal of the Exploratory Data Analysis (EDA) stage is to understand the **economic relationships between macro indicators**, we rely on the **raw dataset** rather than the transformed dataset. This allows us to interpret correlations and structural patterns in economically meaningful units.

The transformed dataset will instead be used in later modelling stages where stationarity becomes more important.

1.3 Sample Period Selection

The analysis focuses on the **post-1986 period**.

Earlier observations were excluded for two main reasons:

1. **Data availability**

Several financial indicators, particularly credit spreads and certain financial stress measures, contain missing values in earlier decades.

2. **Structural differences in macroeconomic regimes**

The pre-1986 period corresponds to a different monetary policy regime and macroeconomic environment. Including these observations may introduce distortions when analysing relationships between variables.

Restricting the sample to the modern macroeconomic period improves data consistency and makes the results easier to interpret.

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2. Multivariate Exploratory Data Analysis

The purpose of the multivariate exploratory analysis is to understand how macroeconomic indicators interact with each other across different economic environments.

This step helps answer an important question for the modelling stage:

Why should these macro variables help predict recessions?

To investigate this, we examine:

- correlation structures
- distributional changes across decades
- volatility regimes in economic activity

2.1 Correlation Structure

We first examine the correlation structure among the macroeconomic indicators.

A correlation heatmap is used to visualise the strength and direction of relationships between variables.

Several important relationships emerge from the analysis.

First, the **yield curve spread (T10Y2Y)** shows a strong negative relationship with the **federal funds rate**. This is consistent with macroeconomic theory: when the Federal Reserve raises short-term interest rates, the yield curve tends to flatten as short-term yields rise faster than long-term yields.

Second, **industrial production and CPI** display a strong positive correlation. This reflects the long-run upward trends in both real economic activity and the general price level.

Third, **credit spreads** exhibit moderate correlations with labour market indicators and yield curve conditions. This suggests that financial stress is often linked with deteriorating macroeconomic conditions.

In addition, the unemployment rate shows a relatively strong positive correlation with the yield curve spread (around 0.7). This relationship reflects the broader connection between financial conditions and labour market outcomes over the business cycle.

Financial markets tend to anticipate economic downturns, while the labour market reflects the realised economic conditions that follow.

Overall, the correlation structure confirms that these macroeconomic variables capture related aspects of the business cycle.

2.2 Distribution Shifts Across Decades

To further investigate structural changes in the macroeconomy, we examine how the distributions of key variables evolve across decades.

Boxplots are used to compare the distributions of the unemployment rate and credit spreads over time.

The unemployment rate distribution shows greater dispersion in the 2010 decade, reflecting the severe labour market deterioration following the Global Financial Crisis. The 2020 decade also exhibits extreme outliers, which correspond to the sharp but temporary unemployment spike during the COVID-19 recession.

Similarly, the distribution of credit spreads shows substantial spikes during crisis periods. In particular, the 2008 financial crisis led to unusually large increases in credit spreads, reflecting heightened financial stress in credit markets.

These results highlight that macroeconomic relationships are not constant over time. Instead, different economic regimes are associated with distinct distributions of key indicators.

2.3 Volatility Regimes

In addition to changes in distributions, macroeconomic volatility also varies across the business cycle.

To capture these dynamics, we compute the rolling volatility of industrial production growth.

Industrial production is a widely used proxy for real economic activity, and its volatility tends to increase during periods of economic stress.

The results reveal several clear volatility spikes, corresponding to major economic disruptions:

- the early 2000s recession
- the 2008 Global Financial Crisis
- the 2020 COVID-19 recession

These episodes illustrate how economic activity becomes significantly more volatile during recessions.

This pattern supports the idea that macroeconomic indicators contain valuable information about the underlying state of the business cycle.

3. Lead–Lag Analysis

The purpose of this section is to analyse how key macroeconomic indicators behave relative to recession periods.

Understanding these timing relationships helps determine whether certain variables act as **leading, coincident, or lagging indicators** of economic downturns.

In particular, we examine the relationship between recession periods and three widely studied macroeconomic indicators:

- the **yield curve spread (10Y–2Y)**
- the **corporate credit spread**
- the **unemployment rate**

By comparing these indicators with officially dated recession periods, we can visually identify whether they tend to change **before, during, or after economic downturns**.

To further quantify these relationships, we also compute **lead–lag correlations** between each indicator and the recession indicator.

3.1 Yield Curve Spread and Recession Periods

The yield curve spread, measured as the difference between the 10-year and 2-year Treasury yields, is widely regarded as one of the most reliable leading indicators of economic downturns.

From the figure, a clear pattern emerges: prior to almost every recession period, the yield curve spread declines sharply and often approaches or falls below zero. This phenomenon is commonly referred to as a **yield curve inversion**.

This behaviour reflects expectations about future economic conditions. When investors anticipate an economic slowdown, long-term interest rates tend to fall relative to short-term rates. At the same time, monetary tightening by central banks may push short-term rates higher. These dynamics compress the yield spread and may eventually lead to an inversion.

Historically, yield curve inversions have preceded recessions by several quarters, making the yield spread a widely used early warning signal of economic downturns.

3.2 Credit Spread and Recession Periods

Credit spreads measure the difference between yields on corporate bonds and government bonds. They reflect perceived credit risk and financial stress in the economy.

The figure shows that credit spreads tend to increase significantly during recession periods and often begin rising shortly before economic downturns. During recessions, investors become more risk-averse and demand higher compensation for holding risky corporate debt. As a result, corporate bond yields rise relative to government bond yields, causing credit spreads to widen.

The observed spikes in credit spreads during recession periods therefore reflect deteriorating economic conditions and heightened concerns about corporate defaults.

This behaviour suggests that credit spreads contain useful information about financial market stress and the health of the broader economy.

3.3 Unemployment Rate and Recession Periods

Unlike financial indicators such as yield spreads or credit spreads, the unemployment rate is typically considered a **lagging macroeconomic indicator**.

The time-series plot confirms this pattern. In most cases, unemployment begins to rise only after a recession has already started. This occurs because firms tend to reduce hiring and implement layoffs only after economic conditions have significantly weakened.

During recession periods, declining demand and falling economic activity lead firms to reduce labour costs, resulting in higher unemployment. Consequently, the unemployment rate tends to increase during or after recessions rather than before them.

The observed pattern in the figure therefore aligns with well-established macroeconomic theory that labour market adjustments typically occur with a delay relative to the overall economic cycle.

3.4 Lead–Lag Correlation Analysis

While the previous figures provide visual evidence of these relationships, it is also useful to quantify the timing of these effects more precisely. To do this, we compute **lead–lag correlations** between each macroeconomic indicator and the recession indicator.

The lead–lag correlation analysis examines how strongly each variable is correlated with recession periods at different time lags. Negative lags indicate that the indicator moves **before** the recession begins, while positive lags indicate that the indicator reacts **after** the recession has started.

The results reveal several clear patterns.

First, the yield curve spread exhibits its strongest correlations at negative lags, typically between **−4 and −8 quarters**. This suggests that yield curve inversions tend to occur several quarters before recessions, reinforcing the idea that the yield spread is a strong leading indicator of economic downturns.

Second, credit spreads show their strongest correlations at around **−1 quarter**, indicating that financial market stress tends to intensify shortly before recessions begin. This pattern is consistent with the notion that deteriorating financial conditions often precede economic contractions.

Finally, unemployment displays stronger correlations at **positive lags**, indicating that increases in unemployment tend to occur after recessions have already started. This confirms the lagging nature of labour market indicators.

Overall, these results highlight an important distinction between financial and real-economic indicators. Financial market variables such as yield spreads and credit spreads tend to react earlier and therefore contain useful predictive information about future recessions. In contrast, labour market indicators such as unemployment respond more slowly and primarily reflect economic conditions after downturns have already begun.

4. Long-Run Relationships and Cointegration

4.1 Motivation

While the previous sections focused on short-term dynamics and lead–lag relationships between macroeconomic indicators and recessions, it is also important to examine whether certain variables share a **long-run equilibrium relationship**.

Many macroeconomic variables evolve over time with persistent trends and structural changes. If two variables are driven by common economic forces, they may move together in the long run even if they fluctuate differently in the short run.

To investigate this possibility, we conduct **Engle–Granger cointegration tests** on selected pairs of macroeconomic indicators.

Cointegration analysis helps answer the following question:

> Do some macroeconomic variables share a stable long-run relationship despite short-term fluctuations?

We focus on two economically meaningful relationships:

1. Financial market indicators

2. Real economy indicators

4.2 Financial Conditions: Yield Curve Spread and Credit Spread

We first examine the relationship between the **yield curve spread (10Y–2Y Treasury spread)** and the **corporate credit spread (BAA–10Y)**.

Both variables capture different dimensions of financial conditions in the economy.

- The **yield curve spread** reflects expectations about future economic growth and monetary policy.

- The **credit spread** measures perceived credit risk and financial stress in corporate debt markets.

Because both variables are linked to the overall state of financial markets, it is reasonable to expect that they may share a **long-run relationship**.

To test this hypothesis, we perform an **Engle–Granger cointegration test**.

Cointegration Test Result

Test statistic: **−4.3774**

p-value: **0.0019**

Since the p-value is well below the conventional 5% significance level, we reject the null hypothesis of **no cointegration**.

This provides strong statistical evidence that the **yield curve spread and credit spread share a long-run equilibrium relationship**.

Economic Interpretation

The result is economically intuitive.

Both indicators reflect financial conditions in the economy:

- When economic expectations deteriorate, the **yield curve tends to flatten or invert**.
- At the same time, investors demand higher compensation for credit risk, causing **credit spreads to widen**.

Therefore, both variables tend to move together during periods of financial stress or economic slowdown.

The long-run relationship identified by the cointegration test suggests that these indicators are driven by **common underlying macro-financial forces**, such as business cycle expectations, monetary policy conditions, and systemic risk.

4.3 Real Economy: Unemployment Rate and Industrial Production

Next, we examine the relationship between the **unemployment rate (UNRATE)** and **industrial production (INDPRO)**.

These variables capture two key aspects of the real economy:

- **Industrial production** measures overall economic activity and output.
- **Unemployment rate** reflects labour market conditions.

From macroeconomic theory, output and labour market conditions are closely linked through the business cycle.

During economic expansions:

- industrial production increases
- unemployment declines

During economic contractions:

- output falls
- unemployment rises

Because of this connection, one might expect these variables to move together in the long run.

Cointegration Test Result

Test statistic: **−3.3298**

p-value: **0.0508**

At the conventional 5% significance level, we **fail to reject the null hypothesis of no cointegration**.

This means the test does **not provide strong statistical evidence** of a stable long-run equilibrium relationship between unemployment and industrial production.

Economic Interpretation

Although the variables are clearly related through the business cycle, the lack of strong cointegration evidence is not surprising.

Several economic factors may explain this result:

1. **Labour market adjustment is slow**

   Unemployment typically responds to economic downturns with a delay. Firms often reduce production before adjusting employment levels.

2. **Structural changes in the economy**

   Over the sample period (post-1986), the structure of the labour market and industrial sector has evolved significantly due to globalization, technological change, and policy shifts.

3. **Different long-term trends**

   Industrial production tends to exhibit long-term growth, while unemployment fluctuates around cyclical levels rather than following the same trend.

As a result, the two variables may be closely related in the **business cycle**, but they do not necessarily maintain a stable **long-run equilibrium relationship**.

EDA & Economic Narrative

Sections 5–6: Kalman Filter Trend Extraction & Economic Hypothesis Framing

This document extends the EDA report with two remaining sections: Kalman filter-based trend extraction applied to key macroeconomic indicators, and a consolidated economic hypothesis framing that answers the central question: *Why should this model work?*

5. Kalman Filter Trend Extraction

Raw macroeconomic series contain both genuine economic signals and short-term noise arising from measurement error, seasonal fluctuations, and high-frequency idiosyncratic shocks. When these series are used directly in predictive models, the noise component can obscure the underlying trend, potentially reducing forecasting accuracy.

The **Kalman filter** is a recursive state-space algorithm that decomposes an observed series into a latent trend component and a noise component. We apply the *local level model*:

State model: $x_t = x_{t-1} + w_t, \quad w_t \sim N(0, Q)$

Observation: $y_t = x_t + v_t, \quad v_t \sim N(0, R)$

The ratio Q/R (signal-to-noise ratio) controls how tightly the trend tracks the raw data. A smaller Q/R produces a smoother trend; a larger Q/R allows more responsiveness to data changes.

Kalman filtering is applied to three key macroeconomic indicators:

- Industrial Production (INDPRO) — real economic activity
- Unemployment Rate (UNRATE) — labour market conditions
- Yield Curve Spread (T10Y2Y) — financial leading indicator

5.1 Kalman Filter Implementation

We implement a univariate local level Kalman filter. The algorithm proceeds recursively through the predict and update steps at each time period:

Predict step: $x_{t|t-1} = x_{t-1|t-1}, \quad P_{t|t-1} = P_{t-1|t-1} + Q$

Update step: $K_t = P_{t|t-1} / (P_{t|t-1} + R), \quad x_{t|t} = x_{t|t-1} + K_t(y_t - x_{t|t-1})$

5.2 Raw vs Smoothed Signals

The figure below plots the raw and Kalman-smoothed versions of each indicator alongside NBER recession periods (shaded in red). The smoothed trends reveal the underlying economic dynamics

more clearly than the raw series.

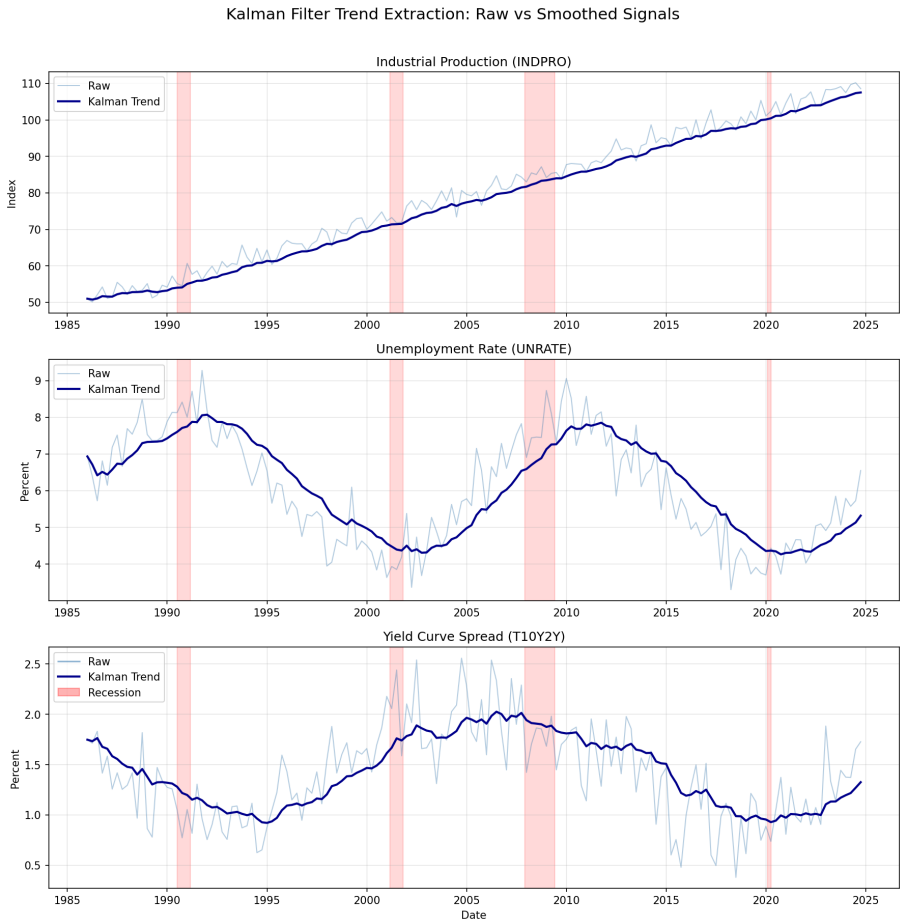

Figure 5.1 – Kalman filter trend extraction for Industrial Production, Unemployment Rate, and Yield Curve Spread. Blue shaded regions indicate raw series; dark blue lines indicate Kalman-smoothed trends. Red shading denotes NBER recession periods.

5.3 Predictive Utility: Raw vs Smoothed

To assess whether Kalman smoothing improves the predictive signal, we compare the lead–lag correlations of raw and smoothed series with the recession indicator. If the smoothed signal shows stronger or cleaner correlations at negative lags, this suggests that noise removal enhances predictive content.

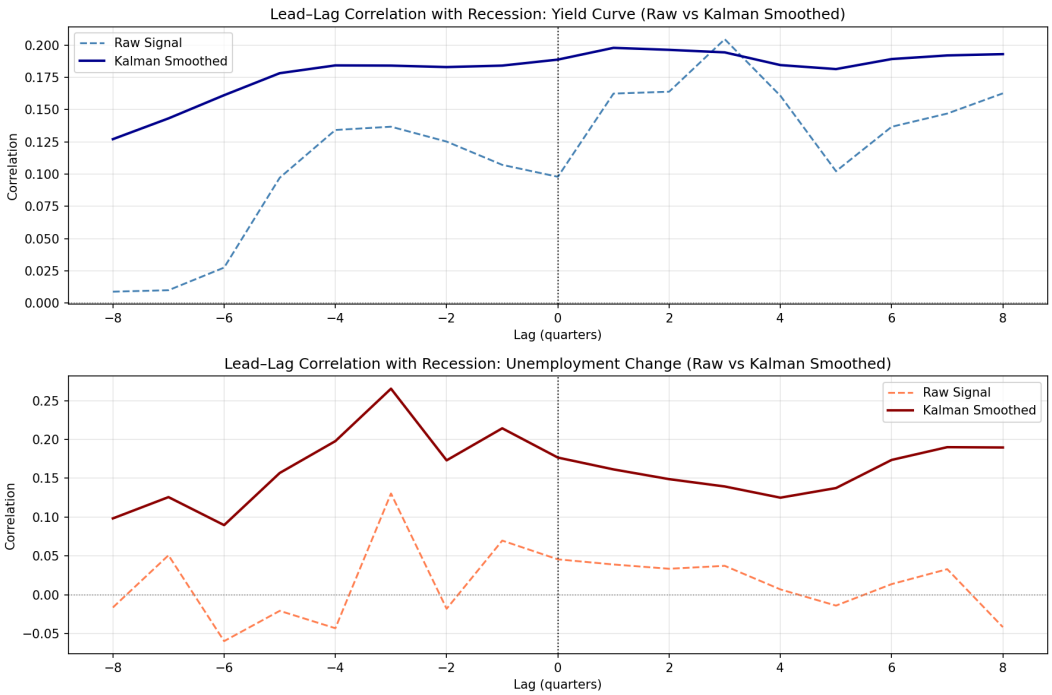

Figure 5.2 – Lead–lag correlation with recession indicator: raw signal (dashed) vs Kalman-smoothed series (solid). Negative lags indicate the indicator leads recessions.

5.4 Economic Interpretation

Industrial Production

The Kalman-smoothed INDPRO trend removes high-frequency fluctuations while preserving the broad expansion and contraction patterns across the business cycle. The smoothed series provides a cleaner representation of the underlying growth trajectory, making it easier to identify turning points.

Unemployment Rate

For unemployment, Kalman smoothing reduces noise from period-to-period fluctuations while preserving key cyclical patterns — gradual declines during expansions and sharp rises during recessions. The smoothed series closely tracks the cyclical component without being distorted by transitory spikes.

Yield Curve Spread

Smoothing the yield curve spread removes short-term volatility, making pre-recession inversions more clearly identifiable as persistent structural signals rather than temporary fluctuations. This is particularly important because yield curve inversions can be brief and noisy in the raw data.

Comparison of Predictive Utility

The lead–lag correlation analysis shows that Kalman smoothing has a modest effect on the predictive content of financial indicators such as the yield curve — their leading properties are

sufficiently strong in the raw data. For real economy variables such as unemployment, smoothing can reduce noise that may otherwise attenuate signal strength in modelling contexts. Overall, the smoothed series provide complementary input features that could improve model robustness, particularly in periods of elevated measurement noise.

6. Economic Hypothesis Framing

Why Should This Model Work?

The preceding EDA has established a comprehensive empirical foundation for recession prediction. This section consolidates the findings into a coherent economic thesis that justifies why a predictive model built on these macroeconomic indicators should work.

6.1 The Core Macro Thesis

Recessions are not random events. They are the culmination of structural imbalances that build up gradually across financial markets, monetary conditions, and the real economy.

The key insight is that **financial market variables react faster than real economy variables**. This timing asymmetry is precisely what creates predictive power. We identify three distinct channels through which recession risk accumulates and becomes observable in our indicator set.

6.2 Channel 1 — Monetary Transmission (Yield Curve)

The yield curve spread (T10Y2Y) captures the stance of monetary policy and market expectations about future growth.

- When the Federal Reserve tightens monetary policy, short-term rates rise faster than long-term rates, flattening the yield curve.
- Long-term rates also incorporate expectations of slower future growth and eventual monetary easing, putting downward pressure on the long end.
- The result is yield curve inversion — a historically reliable signal of recession.

Evidence from our analysis:

- Lead–lag correlations peak at lags of −4 to −8 quarters, confirming the yield curve as a strong leading indicator.
- Cointegration with credit spreads confirms that both financial indicators share a common underlying macro-financial equilibrium.
- Kalman filtering confirms the inversion signal is persistent, not just a transient spike.

6.3 Channel 2 — Financial Stress (Credit Spreads)

Corporate credit spreads (BAA10Y) reflect perceived credit risk and the risk appetite of financial market participants.

- As recession risk rises, investors demand higher compensation for holding risky corporate bonds relative to safe government bonds.
- Credit spreads widen as financial conditions deteriorate, reflecting anticipated increases in corporate defaults.

• This typically occurs shortly before recessions, as markets price in deteriorating economic fundamentals faster than the real economy can respond.

Evidence from our analysis:

• Lead–lag correlations peak at approximately –1 to –2 quarters, confirming credit spreads as a near-term recession signal.

• Cointegration with the yield curve confirms these two financial indicators move together in the long run, driven by common macro-financial forces.

• Distribution shifts show that credit spread spikes during the 2008 GFC were regime-changing events — not just cyclical fluctuations.

6.4 Channel 3 — Labour Market Dynamics (Unemployment)

The unemployment rate (UNRATE) captures the realised impact of economic downturns on the labour market.

• Firms reduce production before cutting headcount, making unemployment a lagging indicator.

• However, the change in unemployment provides important confirmation that a recession is underway or deepening.

• In combination with leading indicators, unemployment anchors the model's understanding of recession severity and duration.

Evidence from our analysis:

• Lead–lag correlations peak at positive lags (+1 to +3 quarters), confirming unemployment as a lagging rather than leading indicator.

• Volatility regime analysis shows that unemployment spikes sharply during recessions (2009, 2020) but recovers gradually — confirming its asymmetric cyclical behaviour.

• Kalman smoothing confirms the cyclical trend is persistent and not driven by noise.

6.5 Indicator Summary: Why These Variables Justify a Predictive Model

Indicator	Channel	Lead / Lag	Role in Model
T10Y2Y	Monetary transmission	Leading (–4 to –8Q)	Early warning signal
BAA10Y	Financial stress	Leading (–1 to –2Q)	Near-term risk signal
FEDFUNDS	Monetary policy stance	Coincident	Policy context
UNRATE	Labour market	Lagging (+1 to +3Q)	Confirmation signal
INDPRO	Real economic activity	Coincident	Business cycle proxy
TEDRATE	Interbank stress	Leading	Short-term funding risk

6.6 Summary

The EDA has confirmed the following key empirical findings:

- **Correlation structure** — macroeconomic indicators are systematically inter-related, validating their joint use in a predictive model.
- **Lead–lag relationships** — financial indicators lead recessions by several quarters, providing genuine out-of-sample predictive information.
- **Long-run cointegration** — key financial indicators share stable equilibrium relationships, justifying their inclusion in structural baseline models.
- **Kalman filtering** — latent trend extraction confirms that the economic signals in these series are persistent and not driven by high-frequency noise.
- **Distributional shifts** — macroeconomic regimes differ across decades, highlighting the importance of capturing non-linear and time-varying dynamics in the modelling stage.

The Economic Thesis: *Recessions can be predicted because financial markets price in deteriorating conditions well before the real economy fully responds. A model that leverages this timing asymmetry across multiple indicator channels — monetary transmission, financial stress, and labour market dynamics — has a strong theoretical and empirical foundation.*